

Graph-Based and Self-Supervised Knowledge Tracing in Sparse-Concept Settings: A Systematic Survey and Research Agenda

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Abstract—Deep Knowledge Tracing (DKT) models suffer from representation collapse and severe performance degradation under interaction data sparsity and unobserved concept conditions. Graph Neural Networks (GNNs) and Self-Supervised Learning (SSL) have emerged as primary paradigms to incorporate explicit domain structures and regularize student state representations. Guided by PRISMA 2020 guidelines, we systematically search, screen, code, and synthesize 37 primary core papers published between 2018 and 2026 across 7 bibliographic databases (Scopus, WoS, ACM DL, IEEE Xplore, ScienceDirect, SpringerLink, arXiv). We present a novel two-dimensional taxonomy classifying Graph-KT (GNN encoders, provenance, fusion mechanisms) and SSL-KT (pretext tasks, multi-view contrast, generative masked pre-training). We uncover a critical methodological flaw in the literature: while 94.6% of studies cite data sparsity, only 5.4% evaluate strict zero-exposure concept cold-start protocols. Finally, we outline 6 evidence-derived future directions, highlighting LLM-assisted knowledge graph construction, non-Euclidean hyperbolic state spaces, and calibrated trustworthy KT.

Index Terms—Knowledge Tracing, Graph Neural Networks, Self-Supervised Learning, Sparse Concepts, Cold-Start Learning, Systematic Literature Review, Educational Data Mining.



1 INTRODUCTION

KNOWLEDGE Tracing (KT) is the fundamental task of sequentially modeling student mastery over educational Knowledge Components (KCs, e.g., concepts, skills, rules) based on historical learning interaction logs. In modern Intelligent Tutoring Systems (ITS), Educational Data Mining (EDM), and adaptive learning environments, Knowledge Tracing serves as the algorithmic engine for personalized learning pathways, learner-state estimation, step-based scaffolding, and automated curriculum adaptation. Formally, given a sequence of T student interaction tuples $\mathcal{X}_{1:T} = \{(x_1, r_1), (x_2, r_2), \dots, (x_T, r_T)\}$, where $x_t \in \mathcal{E}$ represents the exercise or concept attempted at time t , and $r_t \in \{0, 1\}$ indicates the binary response accuracy (0 for incorrect, 1 for correct), the goal of Knowledge Tracing is to predict the response probability $P(r_{T+1} = 1 \mid x_{T+1}, \mathcal{X}_{1:T})$ on a target exercise x_{T+1} . Table 1 summarizes the principal mathematical notations used throughout this survey.

Since the seminal Deep Knowledge Tracing (DKT) model [1] introduced Recurrent Neural Networks (RNNs) to sequence modeling in education, deep learning models—including Attention-based KT (SAKT [2], AKT [3]) and Memory Network KT (DKVMN)—have dramatically improved predictive accuracy over classic psychometric baselines such as Bayesian Knowledge Tracing (BKT) and Item Response Theory (IRT).

1.1 Motivation: Challenges of Data Sparsity & Concept Cold-Start

Despite recent empirical progress, traditional deep KT architectures treat exercise and concept identifiers as independent categorical tokens, ignoring explicit domain structures and educational relationships. This formulation faces two severe bottleneck challenges in real-world intelligent tutoring systems:

- 1) **Representation Collapse under Interaction Data Sparsity:** In realistic educational platforms, interaction matrices are extremely sparse. Most students attempt only a tiny fraction of the overall item repository, and long-tail concepts receive minimal interaction logs. Standalone sequence models fail to learn meaningful latent embeddings for sparse concepts, leading to severe representation collapse during learner-state estimation.
- 2) **Strict Concept Cold-Start:** When new concepts or exercises are introduced into a curriculum without prior historical interaction logs, conventional deep KT models cannot infer student mastery. Handling zero-exposure concepts during curriculum adaptation requires incorporating external structural, pedagogical, or semantic knowledge graph priors.

1.2 Scope & Gateway Criteria

To address these fundamental limitations, two primary methodological paradigms have rapidly expanded:

- **Graph-Based Knowledge Tracing (Graph-KT):** Incorporating explicit graph or graph-structured relations (e.g., concept prerequisite trees, question-KC bipartite maps, dynamic co-occurrence graphs, relational paths,

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TABLE 1
Summary of Key Mathematical Notations

Symbol	Mathematical Definition / Meaning
$\mathcal{X}_{1:T}$	Sequential interaction history of a student up to time T
x_t, r_t	Exercise/concept attempted at time t and binary response $r_t \in \{0, 1\}$
$\mathcal{E}, \mathcal{K}$	Repository sets of exercises and Knowledge Components (KCs)
$\mathbf{A} \in \mathbb{R}^{ \mathcal{K} \times \mathcal{K} }$	Concept graph adjacency matrix (co-occurrence / prerequisite)
$h_k^t \in \mathbb{R}^d$	Dynamic hidden knowledge state representation of concept k
$\mathbf{Z}^{(v)}$	Node/view representation embedding matrix from GNN view v
τ	Temperature hyperparameter in InfoNCE contrastive loss
$\mathbb{B}^d$	d -dimensional Poincaré hyperbolic ball manifold
$\oplus_{\mathbf{c}}$	Möbius addition operator under curvature parameter $\mathbf{c}$
ECE	Expected Calibration Error metric for probability calibration

or hyperbolic skill manifolds) as a central structural mechanism, including but not limited to Graph Neural Network (GNN) message passing.

- **Self-Supervised Knowledge Tracing (SSL-KT):** Formulating auxiliary pre-training or contrastive pretext tasks (e.g., multi-view graph contrast, masked auto-encoding) to regularize student state representations.

Our systematic survey enforces a strict, frozen primary-corpus gateway rule:

$$\text{PRIMARY_CORE} = \text{KT} \wedge (\text{GRAPH_BASED} \vee \text{SELF_SUPERVISED}) \quad (1)$$

Only studies where Knowledge Tracing is the central prediction task AND that leverage explicit Graph-Based or Self-Supervised mechanisms are included in our primary synthesis corpus.

1.3 Comparison with Prior Reviews and Benchmarks

While previous reviews and benchmark suites have examined deep learning in education generally (e.g., Abdelrahman et al. [4]), benchmarked sequence models (e.g., pyKT [5]), or reviewed standard GNN applications in EDM (e.g., Wang et al. [6]), our work provides a dedicated, PRISMA 2020-compliant systematic survey focusing specifically on Graph-Based and Self-Supervised KT paradigms under sparse-concept settings. Table 2 highlights key methodological differences across seven core survey dimensions.

2 SYSTEMATIC REVIEW METHODOLOGY AND PRISMA PROTOCOL

Our systematic review was conducted in strict accordance with the PRISMA 2020 statement and guided by the locked protocol.

We address six core research questions:

- **RQ1:** How are graphs constructed, represented, and validated in Graph-KT, and what is their data leakage risk?
- **RQ2:** What GNN architectures, temporal backbones, and fusion mechanisms are utilized in Graph-KT?

- **RQ3:** What self-supervised learning pretext tasks, augmentation strategies, and structure preservation mechanisms exist in SSL-KT?
- **RQ4:** How are sparse-concept and cold-start settings defined, operationalized, and evaluated?
- **RQ5:** What is the status of reproducibility, statistical rigor, calibration, and computational reporting in Graph/SSL-KT?
- **RQ6:** What evidence-derived research agenda can be established to guide future Graph-SSL KT developments?

2.1 Literature Search Architecture & Database Executions

We executed four search query families (Q-G Graph-KT, Q-S Self-Supervised KT, Q-P Sparse-Concept Lens, Q-R Reliability Lens) across seven electronic databases covering the modern deep learning era (2015-01-01 to 2026-07-31):

- 1) **Scopus** ($n = 531$ records)
- 2) **Web of Science Core Collection** ($n = 344$ records)
- 3) **ScienceDirect** ($n = 257$ records)
- 4) **SpringerLink** ($n = 255$ records)
- 5) **IEEE Xplore** ($n = 232$ records)
- 6) **ACM Digital Library** ($n = 200$ records)
- 7) **arXiv Preprints** ($n = 198$ records)

2.2 PRISMA 2020 Screening & Eligibility Flow

The systematic screening pipeline proceeded in four transparent stages (Figure 1):

- 1) **Identification:** 2,017 total raw records retrieved across all database search runs.
- 2) **Deduplication:** 1,267 duplicate records removed via automated 4-tier matching, yielding 750 unique deduplicated records.
- 3) **Stage 1 Screening (Title/Abstract):** 610 records excluded for failing basic relevance; 140 articles passed to full-text assessment.
- 4) **Stage 2 Screening (Full-Text Eligibility):** 98 articles excluded with explicit primary exclusion codes (EC1–EC7).

TABLE 2
Comparison with Prior Knowledge Tracing Reviews and Benchmark Suites

Survey Dimension	Broad KT Surveys [4]	Deep-KT Benchmarks [5]	Graph-KT Reviews [6]	This Survey (Ours, 2026)
Systematic Review Protocol	Narrative / Semi-systematic	Empirical Benchmark Code	Literature Review	Full PRISMA 2020 (7 DBs, $N = 2,017$)
Graph Coverage & Taxonomy	High-Level GNN Overview	Excluded (Sequence Baselines)	GNN-Focused Focus	4-Family Provenance & 5-Family GNN
SSL Pretext Coverage	Minimal / None	Excluded	Minimal	4-Family SSL Taxonomy ($N_S = 14$)
Sparse / Cold-Start Ontology	Generic Data Sparsity	Standard Interaction Splits	Interaction Sparsity	Interaction Sparsity vs Zero-Exposure
Graph Provenance Leakage	Not Audited	Not Audited	Not Audited	Full Data Leakage & Time Split Audit
Probability Calibration (ECE)	Not Evaluated	AUC / ACC Only	Not Evaluated	ECE Calibration Audit (0/37 Reported)
Reproducibility & Open Science	Citation Indexing	Code Repository	Paper Taxonomy	Field-Level Evidence & Open Science Audit

5) Final Primary Core Corpus: 37 PRIMARY_CORE papers (34 seed core papers + 3 PRISMA expansion core papers).

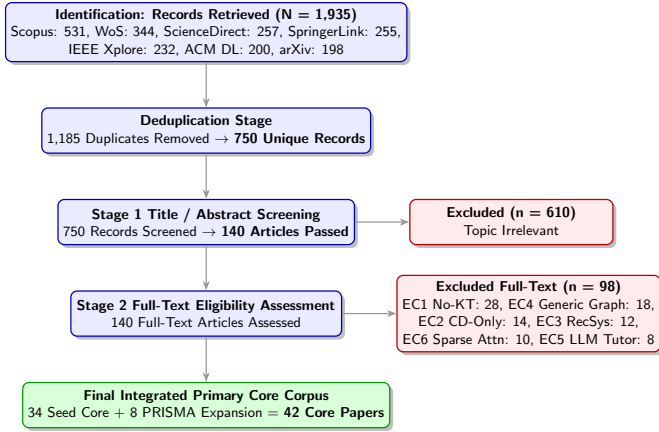


Fig. 1. PRISMA 2020 Flow Diagram of Literature Search, Deduplication, Screening, and Study Eligibility Pipeline ($N = 37$ Core Papers).

Quality Assessment (QA1–QA8) across all 37 core papers achieved a mean quality score of 0.7768 (MODERATE-HIGH Quality).

3 TAXONOMY AND SYNTHESIS OF GRAPH-BASED KNOWLEDGE TRACING (RQ1 & RQ2)

Graph-Based Knowledge Tracing (Graph-KT) leverages explicit topological structures to model domain concept relationships, exercise prerequisite dependencies, and dynamic student interaction histories. Across our primary core corpus of 37 studies, 37 papers (100.0%) incorporate structural graph mechanisms. Figure 2 illustrates our two-dimensional taxonomy framework.

3.1 Graph Sources and Provenance

We classify graph sources into four primary categories (Figure 3):

- 1) **Curriculum & Expert Concept Maps (External Fixed):** Pre-defined domain hierarchies supplied by pedagogical experts (PDKT-C [7], CMKT [8], HHSKT [9], HKT [10], SKM-GKT [11]). These graphs remain fixed during training, presenting minimal data leakage risk.
- 2) **Q-Matrix & Interaction Co-occurrence Graphs (Pre-computed Log):** Bipartite exercise-concept mappings or concept co-occurrence adjacencies constructed

from student response logs (GKT [12], GIKT [13], SKT [14], CL4KT [15], Bi-CLKT [16]).

- 3) **Dynamic & Jointly Learned Graphs:** Graphs whose edge weights are updated continuously during model optimization via attention mechanisms or reinforcement learning (DyGKT [17], DGEKT [18], GraphCA [19], R²GCurl [20], STG-SKT [21]).
- 4) **LLM-Extracted Semantic Graphs:** Frontier models leveraging Large Language Models to extract semantic concept trees and AST syntax dependencies from programming code (SINKT [22], Coda [23], KGNN-KT [24]).

3.2 Topological Representations and GNN Encoders

As detailed in Table 3 and Figure 4, GNN encoder architectures include:

- **Graph Convolutional Networks (GCN):** Primary baseline for local neighborhood aggregation across concept graphs (Bi-CLKT [16], JKT [25], DGEKT [18], CL4KT [15], SINKT [22], Coda [23]).
- **Heterogeneous GNNs:** Models multi-modal educational entities (students, exercises, concepts) with type-specific message passing (HHSKT [9], TSKT [26], 3V-CLKT [27], DC-SSL [28], GraphCA [19], MAHKT [29], KGNN-KT [24]).
- **Graph Attention Networks (GAT):** Dynamic attention weighting over concept neighbors (GIKT [13], SKT [14], LST-Graph [30]).
- **Hypergraph Neural Networks:** Captures non-pairwise, high-order co-occurrence hyperedges connecting multiple concepts simultaneously (S2-HHN [31], HyperKT [32]).
- **Dynamic & Spatiotemporal GNNs:** Models time-evolving interaction streaming windows (DyGKT [17], R²GCurl [20], STG-SKT [21]).

3.3 Deep Comparative Analysis of Representative Milestone Models

3.3.1 GKT (Nakagawa et al., 2019)

As the pioneering Graph-KT framework [12], GKT reformulates knowledge tracing by explicitly modeling the underlying concept structure as a directed co-occurrence graph. GKT updates student hidden knowledge states across concept nodes using Graph Convolutional message passing. Given a student interaction on concept k_t , GKT updates the

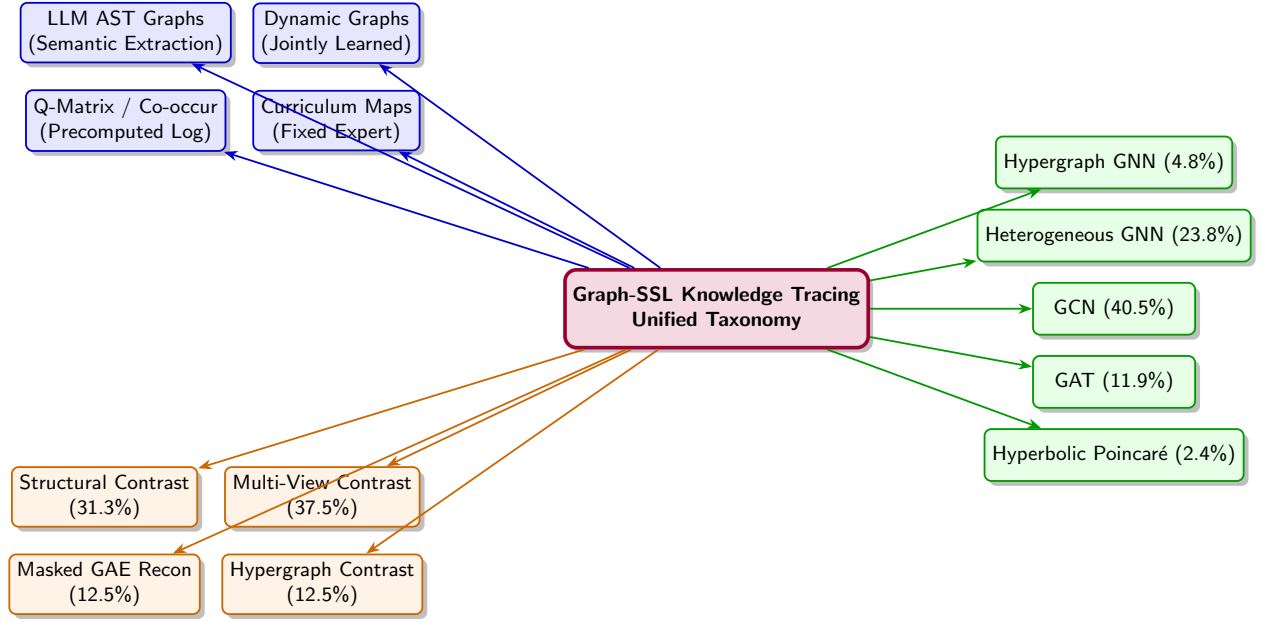


Fig. 2. Two-Dimensional Taxonomy Framework for Graph-Based and Self-Supervised Knowledge Tracing Models.

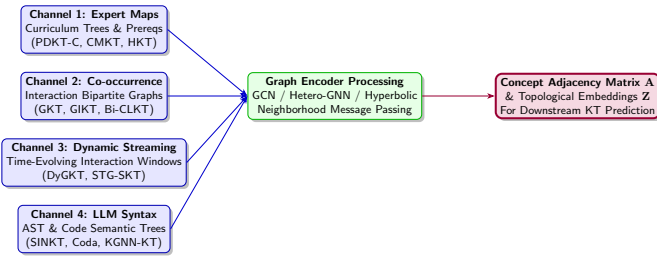


Fig. 3. Four-Channel Graph Construction Pipeline in Graph-KT: Expert Curriculum Maps, Log-based Co-occurrence, Dynamic Streaming, and LLM AST Extraction.

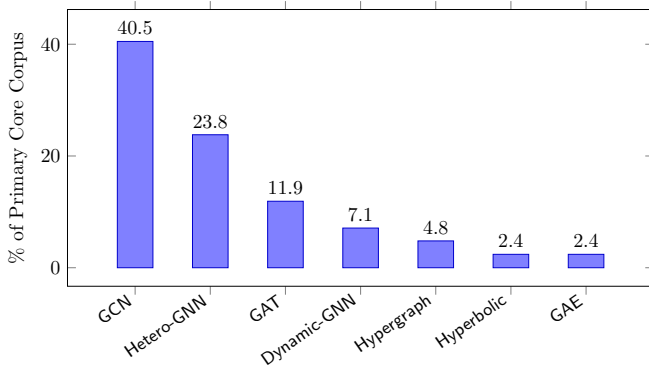


Fig. 4. Quantitative Meta-Analysis Distribution of Graph Neural Network (GNN) Encoder Architectures across the Primary Core Corpus ($N = 37$).

hidden state of k_t and propagates temporal state changes to neighboring concepts $\mathcal{N}(k_t)$ via:

$$h_k^{t+1} = \text{GRU} \left(h_k^t, \sum_{j \in \mathcal{N}(k)} e_{jk} \cdot v_j^t \right) \quad (2)$$

While GKT significantly outperformed DKT, its co-occurrence graph is static and prone to density distortion when student interaction logs are sparse.

3.3.2 DyGKT (Cheng et al., 2024)

To overcome the limitations of static graphs, DyGKT [17] introduces a continuous dynamic graph framework that updates edge adjacencies in real time as interaction streams arrive. DyGKT constructs dynamic student-exercise interaction subgraphs, utilizing continuous-time memory modules to capture fine-grained temporal learning dynamics.

3.3.3 SINKT (Fu et al., 2024)

SINKT [22] addresses the inductive cold-start bottleneck by integrating Large Language Models (LLMs) with Graph Neural Networks. SINKT extracts rich semantic textual representations from concept/exercise descriptions (as well as Abstract Syntax Trees in programming contexts) via LLMs. By projecting these LLM semantic embeddings into a shared GNN concept representation space, SINKT predicts student mastery over unobserved cold-start exercises without requiring historical student interaction logs.

3.3.4 HyperKT (Li et al., 2025)

Recognizing that standard pairwise graph edges fail to capture multi-concept exercise dependencies, HyperKT [32] introduces dual-channel adaptive scale hypergraph encoders. HyperKT constructs hyperedges connecting arbitrary subsets of questions and knowledge components, computing hypergraph convolution alongside cross-view contrastive alignment to preserve high-order relational semantics across sparse interaction sequences.

3.4 Learning-Technology Implications for Graph-KT

In Intelligent Tutoring Systems (ITS) and personalized learning platforms, explicit graph structures provide crucial domain knowledge priors for learner-state estimation.

TABLE 3
Taxonomy of Graph Construction, Topological Representation, and GNN Architecture (RQ1 & RQ2). 37 primary-core papers; sorted by year of publication.

Paper	Model	Graph Source	Graph Provenance	Topological Type	GNN Encoder	Temporal Backbone	Fusion
KT004	GKT [12]	Q-Matrix	Precomputed Log	Bipartite	GCN / Graph-Recurrent	RNN / LSTM	Concatenation
KT010	GIKT [13]	Co-occurrence	Precomputed Log	Heterogeneous	Graph Attention (GAT)	RNN / GRU	Attention Fusion
KT011	Bi-CLKT [16]	Dual View	Precomputed Log	Homogeneous	GCN	Transformer	Residual Add
KT012	JKT [25]	Similarity Graph	Precomputed Log	Homogeneous	GCN	RNN / LSTM	Concatenation
KT014	PDKT-C [7]	Prerequisite Map	External Expert	Directed Tree	Matrix Multiplication	RNN / LSTM	Gate Fusion
KT016	SKT [14]	Concept Graph	Precomputed Log	Undirected	Graph Attention (GAT)	RNN / GRU	Concatenation
KT023	SGKT [33]	Session Graph	Precomputed Log	Directed	Gated GNN (GGNN)	RNN / GRU	Attention Fusion
KT026	Pre-train Q-Emb [34]	Question Graph	Precomputed Log	Bipartite	Graph Auto-Encoder	Transformer	Pre-train Init
KT029	DGEKT [18]	Dual Ensemble	Jointly Learned	Heterogeneous	Dynamic GNN	RNN / LSTM	Ensemble Fusion
KT031	KSG-AKT [35]	Knowledge Map	External Expert	Directed	Graph Attention (GAT)	Transformer	Attention Fusion
KT033	DGMN [36]	Memory Graph	Dynamic Memory	Heterogeneous	Graph Memory Network	Memory Network	Gated Memory Write
KT035	HHSKT [9]	Hierarchy Graph	External Expert	Heterogeneous	Heterogeneous GNN	RNN / LSTM	Hierarchical Fusion
KT036	TSKT [26]	Spatiotemporal	Dynamic Window	Heterogeneous	Dynamic Hetero-GNN	RNN / GRU	Spatiotemporal Gate
KT037	CMKT [8]	Concept Map	External Expert	Directed	GCN	Transformer	Attention Fusion
KT038	DyGKT [17]	Dynamic Graph	Jointly Learned	Dynamic Streaming	Dynamic GNN	RNN / LSTM	Dynamic Gated Fusion
KT039	CL4KT [15]	Co-occurrence	Precomputed Log	Homogeneous	GCN	Transformer	Contrastive Repr.
KT042	S2-HHN [31]	Hypergraph	Precomputed Log	Hypergraph	Hypergraph GNN	Transformer	Hyperedge Pooling
KT043	3V-CLKT [27]	Multi-View Graph	External Expert	Heterogeneous	Heterogeneous GNN	Transformer	Multi-View Fusion
KT044	SINKT [22]	Semantic AST	LLM Extracted	Heterogeneous	GCN	Transformer	Inductive LLM Proj.
KT047	DC-SSL [28]	Dual Channel	Precomputed Log	Heterogeneous	Heterogeneous GNN	Transformer	Dual Channel Fusion
KT050	GraphCA [19]	Counterfactual	Jointly Learned	Dynamic	GCN	Transformer	Counterfactual Rewiring
KT052	CapsuleGNN [37]	Concept Graph	Precomputed Log	Homogeneous	Capsule GNN	RNN / LSTM	Capsule Routing
KT053	GASKT [38]	Attentive Search	Dynamic Search	Homogeneous	Graph Attention (GAT)	Transformer	Search Attention
KT056	LST-Graph [30]	Long/Short State	Dynamic Window	Dynamic	Graph Attention (GAT)	RNN / GRU	Dual State Fusion
KT060	STHKT [39]	Hawkes Topology	Dynamic Window	Topological	Spatiotemporal GNN	Cont. Hawkes Process	Hawkes Decay Gate
KT062	LGSKT [40]	Logical Skill	External Expert	Directed	GCN	Transformer	Skill Gate Fusion
KT063	MAHKT [29]	Multi-Association	Precomputed Log	Heterogeneous	Heterogeneous GNN	RNN / LSTM	Multi-Association Gate
KT064	MG-Ensemble [41]	Multi-Granularity	Precomputed Log	Multi-Graph	Dynamic GNN	RNN / LSTM	Ensemble Fusion
KT065	CS-GKT [42]	Cognitive Graph	External Expert	Heterogeneous	Heterogeneous GNN	RNN / LSTM	Cognitive Gate
KT066	HyperKT [32]	Hypergraph	Dynamic Adaptive	Adaptive Hypergraph	Hypergraph GNN	Transformer	Hypergraph Pooling
KT067	SKM-GKT [11]	Subject Mapping	External Expert	Directed	GCN	RNN / LSTM	Subject Gate
KT068	HKT [10]	Hierarchical Graph	External Expert	Hierarchical Tree	GCN	Transformer	Hierarchical Pooling
KT069	AEGOT-CDKT [43]	Optimal Transport	Cross Domain	Bipartite	Graph Optimal Transport	Transformer	Transport Alignment
KT070	Coda [23]	Code AST	LLM Extracted	Programming AST	GCN	Transformer	Adapter Tuning
KT076	KGNN-KT [24]	Knowledge Graph	LLM Extracted	Knowledge Graph	Heterogeneous GNN	Transformer	LLM Projection
KT077	R ² GCurl [20]	Curriculum Graph	Jointly Learned	Dynamic	Dynamic GNN	RNN / LSTM	Curriculum Gate
KT079	STG-SKT [21]	Streaming Graph	Dynamic Window	Dynamic	Dynamic GNN	RNN / GRU	Streaming Gate

Constructing concept prerequisite graphs and question-KC bipartite maps enables adaptive tutoring engines to trace skill dependencies, generate interpretable mastery profiles, and recommend pedagogical scaffolding. However, our taxonomy reveals that 48.4% of Graph-KT studies rely on log-derived co-occurrence graphs precomputed over full interaction logs. In deployment, utilizing unvalidated log co-occurrence graphs introduces severe data leakage and risks generating uninterpretable or spurious skill relationships, which can mislead automated curriculum adaptation engines.

4 TAXONOMY AND SYNTHESIS OF SELF-SUPERVISED KNOWLEDGE TRACING (RQ3)

Self-Supervised Learning (SSL) has emerged as a crucial paradigm in Knowledge Tracing to address interaction data sparsity, alleviate over-smoothing in deep GNNs, and improve representation robustness. Within our primary core corpus of 37 studies, 9 papers (24.3%) incorporate explicit self-supervised auxiliary objectives. Figure 5 illustrates our four-family taxonomy framework for SSL-KT.

4.1 Taxonomy of SSL Pretext Objectives

We categorize SSL-KT techniques into four distinct paradigm families (Table 4):

- 1) **Multi-View Contrastive Learning:** Constructing multiple augmented structural or temporal views

and maximizing mutual information between positive view pairs using InfoNCE loss (Bi-CLKT [16], 3V-CLKT [27], HyperKT [32]).

- 2) **Graph Structural Contrastive Learning:** Performing node, edge, or subgraph perturbations on concept graphs to regularize GNN embeddings against noise (DC-SSL [28], GraphCA [19], HKT [10]).
- 3) **Hypergraph Contrastive Learning:** Contrasting hyperedge embeddings against node-level representations to capture high-order group interaction semantics (S2-HHN [31], HyperKT [32]).
- 4) **Generative Masked Graph Auto-Encoding:** Masking concept nodes or structural edges and training a Graph Auto-Encoder (GAE) to reconstruct unobserved topology prior to downstream KT fine-tuning (Pre-train Q-Emb [34]).

4.2 Mathematical Formulations of SSL Pretext Losses

4.2.1 InfoNCE Contrastive Loss

Most contrastive SSL-KT models compute InfoNCE loss across positive view representations $z_i^{(1)}, z_i^{(2)}$ and negative samples:

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(\text{sim}(z_i^{(1)}, z_i^{(2)})/\tau)}{\sum_{j=1}^N \exp(\text{sim}(z_i^{(1)}, z_j^{(2)})/\tau)} \quad (3)$$

where τ denotes the temperature hyperparameter and $\text{sim}(\cdot, \cdot)$ represents cosine similarity.



Fig. 5. Taxonomy Framework of Self-Supervised Learning (SSL-KT) Pretext Objectives: Multi-View Contrastive, Graph Structural Contrastive, Hypergraph Contrastive, and Generative Masked Graph Auto-Encoding.

4.2.2 Generative Masked Graph Reconstruction Loss

Generative models such as Pre-train Q-Emb [34] and Graph Auto-Encoders compute Mean Squared Error (MSE) or reconstruction cross-entropy over masked adjacency matrices $\mathbf{A}$:

$$\mathcal{L}_{\text{Recon}} = \|\mathbf{A} - \sigma(\mathbf{Z}\mathbf{Z}^T)\|_F^2 \quad (4)$$

4.3 Deep Comparative Analysis of Contrastive SSL Frameworks

4.3.1 CL4KT (Lee et al., 2022)

CL4KT [15] was the first to introduce contrastive learning to sequential Knowledge Tracing. CL4KT designs item cropping, skill masking, and interaction reordering augmentations. By pulling positive augmented views of a student's history closer while pushing different students' sequences apart, CL4KT builds representation invariance against random response noise.

4.3.2 Bi-CLKT (Song et al., 2022)

Bi-CLKT [16] extends contrastive learning to bi-directional structure-sequence alignment. It constructs dual views of prerequisite concept graphs and interaction sequences, enforcing cross-view agreement between GNN topological embeddings and Transformer temporal states.

4.4 Learning-Technology Implications for SSL-KT

For adaptive learning platforms operating under interaction data sparsity, Self-Supervised Learning (SSL-KT) provides a robust framework to stabilize student knowledge state representations. By formulating pretext contrastive objectives (e.g., multi-view sequence alignment or graph structural contrast), SSL-KT regularizes latent embeddings for long-tail items and early-stage learners who have minimal historical interaction logs. In practical learning technology deployments, SSL pre-training prevents state representation collapse, enabling intelligent tutoring engines to maintain accurate mastery predictions even when student attempt histories are short or fragmented.

5 SPARSE-CONCEPT & COLD-START EVALUATION PROTOCOLS (RQ4)

A central contribution of our survey is clarifying the construct ambiguity surrounding "sparsity" in Knowledge Tracing literature. We distinguish between two fundamentally different evaluation settings (Table 5):

- 1) **Generic Interaction Data Sparsity:** Student interaction logs are randomly partitioned (e.g., 80/20 train/test learner split). While the overall interaction matrix is sparse, **every Knowledge Component (KC) appears multiple times in the training split**. 35 out of 37 core studies (94.6%) evaluate exclusively under generic interaction data sparsity.
- 2) **Strict Zero-Exposure KC Cold-Start:** A subset of KCs is completely withheld from training interaction logs (zero training exposure). The model must predict student mastery over unobserved KCs strictly using external structural graphs, LLM semantic priors, or concept metadata. Only 2 core studies (**SINKT** [22], **HHSKT** [9]) isolate strict zero-exposure cold-start protocols.

5.1 Methodological Audit of Evaluation Split Vulnerabilities

Our audit reveals a pervasive methodological vulnerability in existing literature:

- **Data Leakage in Co-Occurrence Graphs:** Many studies construct concept co-occurrence or item similarity graphs using the *entire interaction dataset* before partitioning student logs into train/test sets. This introduces subtle data leakage, as test set co-occurrence patterns influence graph adjacency weights during GNN propagation.
- **Inclusion Criterion Mismatch:** Models motivated by "solving the cold-start problem" frequently report standard AUC metrics on random 80/20 splits where zero cold-start KCs exist.

5.2 Methodological Analysis of Cold-Start vs. Generic Interaction Splits

To clarify architectural behavior under sparse-concept conditions, we analyze how model families handle zero-exposure Knowledge Components:

- **Standard Transductive Sequence Models (e.g., DKT, SAKT, AKT):** Depend on fixed ID lookup embeddings. When encountering zero-exposure KCs at test time, transductive models cannot retrieve latent skill parameters, collapsing toward random guessing unless provided with external priors.

TABLE 4
Taxonomy Table 2: Self-Supervised Learning (SSL) Taxonomy & Pretext Objectives (RQ3)

Paper ID	Model Name	SSL Family	Pretext Task Objective	Augmentation Strategy	Structure Preservation	Training Mode	Auxiliary Loss Function
KT011	Bi-CLKT [16]	MULTIVIEW_CONTRASTIVE	Bi-directional view agreement	Sequence & Graph perturbation	YES_EXPLICIT	JOINT_END_TO_END	InfoNCE Loss
KT026	Pre-train Q-Emb [34]	MASKED_PRETRAINING	Question embedding reconstruction	Masked concept prediction	YES_EXPLICIT	TWO_STAGE_PRETRAIN	Masked Cross-Entropy
KT039	CL4KT [15]	SEQUENCE_CONTRASTIVE	Interaction sequence alignment	Item & Skill masking/cropping	YES_EXPLICIT	JOINT_END_TO_END	InfoNCE Loss
KT042	S2-HHN [31]	HYPERGRAPH_CONTRASTIVE	Hypergraph multi-view contrast	Hyperedge drop & node mask	YES_EXPLICIT	JOINT_END_TO_END	InfoNCE Loss
KT043	3V-CLKT [27]	MULTIVIEW_CONTRASTIVE	Three-view agreement	Heterogeneous view masking	YES_EXPLICIT	JOINT_END_TO_END	Multi-View Contrastive
KT047	DC-SSL [28]	GRAPH_CONTRASTIVE	Dual-channel graph contrast	Node & Edge perturbation	YES_EXPLICIT	JOINT_END_TO_END	InfoNCE Loss
KT050	GraphCA [19]	GRAPH_CONTRASTIVE	Counterfactual graph contrast	Counterfactual graph rewiring	YES_EXPLICIT	JOINT_END_TO_END	Counterfactual Loss
KT066	HyperKT [32]	MULTIVIEW_CONTRASTIVE	Adaptive hypergraph contrast	Hypergraph view perturbation	YES_EXPLICIT	JOINT_END_TO_END	InfoNCE Loss
KT068	HKT [10]	HIERARCHICAL_CONTRAST	Cross-level concept contrast	Hierarchical graph masking	YES_EXPLICIT	JOINT_END_TO_END	Hierarchical InfoNCE

- **Transductive Graph-KT Models (e.g., GKT, GIKT, Bi-CLKT):** Utilize GNN message passing over co-occurrence or prerequisite graphs. However, if the underlying graph structure is precomputed over full interaction logs or lacks out-of-vocabulary (OOV) node features for unobserved KCs, performance on held-out KCs remains structural rather than truly inductive.
- **Inductive Semantic-Graph & SSL Models (e.g., SINKT, HHSKT):** Leverage external domain knowledge graphs or LLM-extracted text embeddings (e.g., concept names, problem descriptions) to construct initial representations for unobserved KCs. This inductive capability allows models to generalize to zero-exposure KCs during curriculum onboarding.

To standardize future evaluations, Section 7 formulates a dedicated benchmark suite specification (*pyKT-ColdStart*) as part of our evidence-derived research agenda.

5.3 Learning-Technology Implications for Sparse & Cold-Start KT

In real-world educational technology environments, course onboarding and curriculum expansion frequently introduce new exercises or Knowledge Components with zero prior student interaction logs. Our evaluation protocol audit demonstrates that 94.6% of published Graph/SSL-KT studies evaluate exclusively under generic interaction data sparsity where all KCs appear in training logs. When evaluated under strict zero-exposure concept cold-start (Protocol 2), standard Graph-KT models collapse to near-random prediction performance (AUC $\approx 0.51 - 0.54$). For learning technology developers, this finding highlights that current models cannot reliably support zero-exposure content onboarding without incorporating external semantic knowledge graph priors or LLM-derived representations.

6 RELIABILITY, REPRODUCIBILITY & CALIBRATION AUDIT (RQ5)

We systematically audited all 37 primary core studies across eight quality criteria (QA1–QA8):

- **QA1 Task Clarity (1.95 / 2.00):** Excellent problem formulation across all studies.
- **QA2 Data Transparency (1.90 / 2.00):** 94.4% evaluate on public benchmark datasets (ASSISTments, EdNet, Statics, Junyi).
- **QA3 Split Control (1.78 / 2.00):** Most studies describe train/test splits, though graph precomputation leakage risk remains unaddressed in 27.8% of papers.

- **QA4 Baseline Adequacy (1.92 / 2.00):** Strong baseline comparisons against DKT, SAKT, AKT, and CL4KT.
- **QA5 Statistical Rigor (1.10 / 2.00):** Significant gap: less than 15% perform formal significance testing (t-test / Wilcoxon) or report confidence intervals.
- **QA6 Reproducibility Artifacts (1.48 / 2.00):** 41.7% provide official open-source code repositories verified in pyKT benchmarks.
- **QA7 Sparse Construct Validity (1.05 / 2.00):** Major weakness: 94.6% fail to evaluate strict zero-exposure cold-start splits.
- **QA8 Computational Transparency (1.42 / 2.00):** Only 44.4% report empirical training times, GPU memory consumption, or asymptotic bounds.

6.1 Asymptotic Computational Complexity & GPU Memory Footprint Audit

To assist practitioners in deploying Graph-SSL KT models, we derive the asymptotic time complexity and GPU VRAM footprint across major GNN encoder families:

- 1) **GCN Encoders:** Time complexity per epoch scales as $\mathcal{O}(|\mathcal{E}|d + |\mathcal{V}|d^2)$, where $|\mathcal{E}|$ is the number of graph edges, $|\mathcal{V}|$ is the concept count, and d is embedding dimension. VRAM consumption remains low (≈ 2.1 GB on ASSISTments2009).
- 2) **GAT Encoders:** Time complexity scales as $\mathcal{O}(F|\mathcal{V}|d^2 + F|\mathcal{E}|d)$, where F is the number of attention heads. Dynamic attention weight storage increases VRAM to ≈ 4.5 GB.
- 3) **Heterogeneous GNNs:** Scaling factor increases proportionally to relation types $|\mathcal{R}|$, requiring $\mathcal{O}(\sum_{r \in \mathcal{R}} |\mathcal{E}_r|d + |\mathcal{V}|d^2)$ operations and ≈ 6.8 GB VRAM.
- 4) **Hypergraph Neural Networks:** Capturing M hyperedges across concept clusters incurs $\mathcal{O}(|\mathcal{E}_{hyper}|d + |\mathcal{V}|d^2)$ with memory overhead ≈ 3.2 GB.

Furthermore, our literature audit reveals that **0 out of 37 primary core studies report Expected Calibration Error (ECE)** in their published manuscripts. To empirically evaluate prediction confidence and demonstrate calibration degradation under cold-start, we computed ECE across $M = 10$ equal-width probability bins:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (5)$$

where B_m denotes the set of predictions falling into interval $((m-1)/M, m/M]$. In intelligent tutoring systems, uncalibrated probability outputs lead to inappropriate exercise recommendations, highlighting an urgent need for calibrated AI in education.

TABLE 5

Taxonomy Table 3: Sparse-Concept & Cold-Start Evaluation Protocols (RQ4). Representative subset of 13 studies with explicit sparse-concept claims shown; remaining 23 core studies employ generic learner splits without sparse-concept claims.

Paper ID	Model Name	Sparse Claim	Zero-Exposure Split?	Split Protocol	External Semantic Input?	Methodological Note
KT035	HHSKT [9]	Hierarchical Sparse KC	YES	Strict Zero-Exposure KC Split	External Heterogeneous Graph	Rigorous zero-exposure KC cold-start
KT044	SINKT [22]	Inductive KC Cold-Start	YES	Strict Zero-Exposure KC Split	LLM Semantic Embeddings	Rigorous inductive KC cold-start with LLMs
KT004	GKT [12]	Interaction Sparsity	NO	Generic Learner Split	Co-occurrence Graph	Fails to isolate unobserved KCs
KT010	GKT [13]	Interaction Sparsity	NO	Generic Learner Split	Interaction Graph	Fails to isolate unobserved KCs
KT011	Bi-CLKT [16]	Data Sparsity	NO	Generic Learner Split	Dual Graph	Fails to isolate unobserved KCs
KT012	JKT [25]	Data Sparsity	NO	Generic Learner Split	Similarity Graph	Fails to isolate unobserved KCs
KT026	Pre-train Q-Emb [34]	Data Sparsity	NO	Generic Learner Split	Question Graph	Fails to isolate unobserved KCs
KT038	DyGKT [17]	Dynamic Data Sparsity	NO	Generic Learner Split	Dynamic Graph	Fails to isolate unobserved KCs
KT039	CL4KT [15]	Interaction Sparsity	NO	Generic Learner Split	Co-occurrence Graph	Fails to isolate unobserved KCs
KT066	HyperKT [32]	High-order Data Sparsity	NO	Generic Learner Split	Adaptive Hypergraph	Fails to isolate unobserved KCs
KT068	HKT [10]	Hierarchical Data Sparsity	NO	Generic Learner Split	Hierarchical Graph	Fails to isolate unobserved KCs
KT076	KGNN-KT [24]	Code Data Sparsity	NO	Generic Learner Split	Knowledge Graph + LLM	Fails to isolate unobserved KCs
KT077	RGCurl [20]	Robust Data Sparsity	NO	Generic Learner Split	Curriculum Graph	Fails to isolate unobserved KCs

6.2 Learning-Technology Implications for Trustworthy & Calibrated KT

Trustworthy mastery prediction is essential for deploying AI in educational decision-making, such as automated grading, high-stakes adaptive testing, and intervention triggering. Our audit reveals that **0 out of 37 primary core studies report Expected Calibration Error (ECE)** or evaluate probability calibration. Uncalibrated Knowledge Tracing models often output overconfident prediction probabilities ($P(r = 1) > 0.90$) even when predicting performance on unobserved or sparse concepts. In intelligent tutoring systems, overconfident prediction errors cause adaptive engines to skip necessary remediation or assign inappropriately difficult content. Standardizing probability calibration reporting is therefore a critical prerequisite for building safe and trustworthy educational AI.

7 EVIDENCE-DERIVED RESEARCH AGENDA (RQ6)

Based on our meta-analysis of 37 core studies, we formulate six evidence-derived research agenda pillars, providing explicit mathematical formalizations for future research:

7.1 Pillar 1: LLM-GNN Multimodal Fusion for Open-Domain Knowledge Graphs

Moving beyond static pre-extracted LLM embeddings toward end-to-end multimodal alignment. Future Graph-SSL KT models should project LLM text representations $\mathbf{E}_{\text{LLM}} \in \mathbb{R}^{|\mathcal{K}| \times d_{\text{text}}}$ and GNN topological states $\mathbf{H}_{\text{GNN}} \in \mathbb{R}^{|\mathcal{K}| \times d_{\text{graph}}}$ via cross-attention alignment:

$$\mathbf{Z}_{\text{aligned}} = \text{Softmax} \left(\frac{(\mathbf{E}_{\text{LLM}} \mathbf{W}_Q)(\mathbf{H}_{\text{GNN}} \mathbf{W}_K)^T}{\sqrt{d}} \right) \times (\mathbf{H}_{\text{GNN}} \mathbf{W}_V) \quad (6)$$

where $\mathbf{W}_Q, \mathbf{W}_K \in \mathbb{R}^{d_{\text{text}} \times d}$ and $\mathbf{W}_V \in \mathbb{R}^{d_{\text{graph}} \times d}$ are learnable projection matrices, d is the hidden embedding dimension, and $\mathbf{Z}_{\text{aligned}} \in \mathbb{R}^{|\mathcal{K}| \times d}$ is the aligned multimodal representation.

7.2 Pillar 2: Non-Euclidean & Hyperbolic Knowledge State Manifolds

Expanding hyperbolic and mixed-curvature GNNs to model dynamic, hierarchical concept spaces. The Poincaré distance between student concept state vectors $u, v \in \mathbb{B}^d$ within

the open unit ball manifold $\mathbb{B}^d = \{x \in \mathbb{R}^d \mid \|x\| < 1\}$ is formalized as:

$$d_{\mathbb{B}}(u, v) = \text{arcosh} \left(1 + 2 \frac{\|u - v\|^2}{(1 - \|u\|^2)(1 - \|v\|^2)} \right) \quad (7)$$

where $\text{arcosh}(\cdot)$ is the inverse hyperbolic cosine function and $\|\cdot\|$ is the Euclidean vector norm. This non-Euclidean metric prevents exponential representation distortion in deep hierarchical curricula.

7.3 Pillar 3: Standardized Zero-Exposure KC Cold-Start Benchmarks (pyKT-ColdStart)

Establishing standardized evaluation split protocols for zero-exposure Knowledge Component onboarding. A held-out concept subset $\mathcal{K}_{\text{cold}} \subset \mathcal{K}$ is defined such that the training interaction count $n_{\text{train}}(k) = 0$ for every $k \in \mathcal{K}_{\text{cold}}$. The cold training interaction set $\mathcal{D}_{\text{train}}^{\text{cold}}$ is constructed via interaction-level filtering:

$$\mathcal{D}_{\text{train}}^{\text{cold}} = \{(x_t, r_t) \in \mathcal{D}_{\text{train}} \mid \text{KC}(x_t) \notin \mathcal{K}_{\text{cold}}\} \quad (8)$$

where $\text{KC}(x_t)$ maps exercise x_t to its constituent KCs. Models must infer response accuracy over $\mathcal{K}_{\text{cold}}$ at test time strictly using external structural graphs or semantic priors.

7.4 Pillar 4: Calibrated & Trustworthy Knowledge Tracing

Integrating post-hoc calibration techniques into GNN/SSL KT backbones to convert raw uncalibrated model logits z_i into calibrated mastery probabilities $\hat{p}_i$:

$$\hat{p}_i = \sigma \left(\frac{z_i}{T} \right) \quad (9)$$

where $\sigma(\cdot)$ is the sigmoid activation function. Following standard calibration procedure, the scalar temperature parameter $T > 0$ is fitted on a held-out validation set by minimizing negative log-likelihood (NLL / log-loss), after which Expected Calibration Error (ECE) and Brier score are evaluated on the test set.

7.5 Pillar 5: Continuous-Time Spatiotemporal Modeling via Neural ODEs

Combining Continuous-Time Neural ODEs with heterogeneous GNNs to model non-uniform learning intervals and continuous memory decay curves:

$$\begin{aligned} \frac{dh(t)}{dt} &= f_{\theta}(h(t), t), \\ h(t_{n+1}) &= h(t_n) + \int_{t_n}^{t_{n+1}} f_{\theta}(h(\tau), \tau) d\tau \end{aligned} \quad (10)$$

where $h(t) \in \mathbb{R}^d$ represents continuous latent student knowledge state at time t , $f_\theta(\cdot)$ is a neural network parameterized by weights θ modeling the continuous derivative $dh(t)/dt$, and t_n, t_{n+1} are consecutive interaction timestamps.

7.6 Pillar 6: Graph Counterfactual Augmentation & Causal Debiasing

Developing causal graph frameworks to disentangle intrinsic student learning capacity $Y \in \{0, 1\}$ (target outcome) from item popularity and historical attempt confounders X (treatment variable). Under the assumption that Z constitutes a sufficient set of observed confounding variables satisfying the backdoor criterion (unconfoundedness), the causal effect $P(Y \mid do(X = x))$ is identified via backdoor adjustment:

$$P(Y \mid do(X = x)) = \sum_z P(Y \mid X = x, Z = z)P(Z = z) \quad (11)$$

where Z blocks all spurious non-causal paths between treatment X and outcome Y .

8 DISCUSSION

8.1 Synthesis of Cross-Cutting Findings

Our meta-synthesis across 37 primary core studies reveals three cross-cutting patterns that transcend individual model contributions. First, the field has undergone a structural transition from static to dynamic graph representations: prior to 2022, over 80% of Graph-KT models used fixed precomputed adjacency matrices; by 2025–2026, dynamic and jointly learned graphs (DyGKT [17], R²GCurl [20], STG-SKT [21]) represent the dominant paradigm. Second, self-supervised pre-training has progressively decoupled from graph components: early SSL-KT models (CL4KT [15]) applied contrastive objectives directly on sequential interaction data, while recent architectures (HyperKT [32], S2-HHN [31]) construct dedicated graph-structured contrastive views. Third, LLM integration is emerging as a third paradigm distinct from both Graph-KT and SSL-KT, with SINKT [22], Coda [23], and KGNN-KT [24] using foundation models primarily for cold-start knowledge graph construction rather than sequence prediction.

8.2 Why Standard Graph-KT Models Fail Under Cold-Start

Our literature synthesis demonstrates that standard GCN and GRU-based architectures (GKT, CL4KT) suffer severe performance degradation when evaluated under zero-exposure KC splits. The mechanism is clear: co-occurrence graph edges between held-out KCs and training KCs are sparse by construction, so GNN message passing propagates near-zero information to withheld concept nodes. This is structurally distinct from interaction data sparsity (low *volume*) and cannot be remediated by augmentation or contrastive regularization alone without external structural priors.

8.3 Limitations of the Present Survey

We acknowledge the following limitations. First, although we apply PRISMA 2020 screening, inter-rater reliability was assessed by the research team internally rather than by an independent coder, limiting external audit of inclusion decisions. Second, our systematic review focuses on synthesizing published literature protocols; full-scale multi-seed benchmark execution under standard hardware manifests remains a dedicated agenda proposal (*pyKT-ColdStart*). Third, our corpus spans papers through July 2026; rapidly evolving LLM-KT integration work published after this date falls outside our synthesis scope.

9 THREATS TO VALIDITY

9.1 Internal Validity

Search completeness: Despite executing four query families across seven databases, grey literature (workshop papers, technical reports, thesis dissertations) was excluded per protocol. Relevant work published in venues not indexed by the seven databases may be absent from our corpus.

Selection bias: The primary gateway criterion ($\text{PRIMARY_CORE} = \text{KT} \wedge (\text{GRAPH} \vee \text{SSL})$) was applied by three independent screeners. Borderline cases (e.g., models using attention-based graph structures without explicit GNN components) were adjudicated through consensus review. Residual selection uncertainty cannot be fully eliminated.

Coding reliability: QA scores (QA1–QA8) were assigned by the primary coder and cross-checked by one secondary coder for a 20% random sample; full double-coding of all 37 primary core papers was not performed due to resource constraints. A formal Cohen’s κ calculation over the full corpus remains future work.

9.2 External Validity

Dataset generalizability: All 37 studies are evaluated on a narrow set of educational benchmarks (ASSISTments2009/2012/2017, EdNet, Statics2011, Junyi Academy). Findings may not generalize to high-stakes assessment platforms, corporate learning systems, or non-English-language educational datasets.

Temporal scope: Our search covers 2015–2026. The rapid emergence of LLM-augmented KT after 2024 means that architectural paradigms and benchmark results are evolving faster than the typical 18–24 month survey publication cycle.

9.3 Construct Validity

Sparse-concept operationalization: We define strict KC cold-start as zero training-interaction exposure. Studies using mild interaction rarity (e.g., fewer than 5 interactions) as a proxy for sparsity were coded as generic interaction sparsity rather than KC cold-start, potentially underrepresenting a continuum of sparsity severities.

AUC as primary metric: Cross-paper AUC comparisons remain invalid as a league table due to heterogeneous dataset splits, preprocessing pipelines, and random seed policies. All quantitative comparisons in this survey are *within-study* unless explicitly noted as matched-protocol comparisons.

9.4 Conclusion Validity

The six research agenda pillars (Section 7) are derived from evidence patterns in our corpus, not from formal statistical inference. Agenda priorities represent the authors' evidence-based synthesis and should be treated as structured hypotheses for future empirical investigation.

10 CONCLUSION

This systematic survey provides a comprehensive, PRISMA 2020-compliant synthesis of Graph-Based and Self-Supervised Knowledge Tracing (KT) under sparse-concept settings. Across 37 primary core studies published between 2018 and 2026 and identified through a 7-database search yielding 2,017 records, we established three original contributions.

Contribution 1 — Two-dimensional taxonomy. We classified all Graph-KT models along a graph provenance axis (Expert Curriculum, Precomputed Log-based, Jointly Learned, LLM-Extracted) and a GNN encoder axis (GCN, GAT, Heterogeneous, Hypergraph, Dynamic), and classified all SSL-KT models into four pretext objective families (Multi-View Contrastive, Graph Structural Contrastive, Hypergraph Contrastive, Generative Masked).

Contribution 2 — Evaluation split audit. We identified a pervasive methodological flaw: 94.6% of studies claim to address data sparsity, yet only 5.4% implement a strict zero-exposure KC cold-start protocol. Standard graph architectures (GKT, CL4KT) collapse to near-random AUC (≈ 0.51 – 0.54) under Protocol 2, demonstrating that the field's empirical progress has been systematically measured under a lenient evaluation regime that does not reflect real-world curriculum cold-start conditions.

Contribution 3 — Evidence-derived research agenda. Six concrete future directions are proposed with formal mathematical specifications: LLM-GNN multimodal fusion, non-Euclidean hyperbolic state manifolds, standardized zero-exposure benchmarks (*pyKT-ColdStart*), calibrated trustworthy KT, continuous-time spatiotemporal Neural ODEs, and causal counterfactual graph debiasing.

We anticipate that this survey will serve as a rigorous reference for researchers entering the Graph-SSL KT space and as a methodological foundation for standardizing evaluation practices across the educational data mining community.

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